Ramu Rathlavath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.83
Chaitanya Junior College, Balapur	Class XII (XII (TSBIE))	2026	95.90%
Z P P High School, Dindi, Nalgonda	Class X (X (TS SSC Board))	2024	8.50%

EXPERIENCE

IC Package Design Engineer | HCL Technologies, Jigani SEZ, Bangalore (*Ongoing)* (July 2026 - Present)

- Working within the ERS Product Engineering - High Speed Digital Unit (SpTech/Capability) team on IC package design and characterisation for DDR5 products.
- Supporting package design reviews and DFX feedback for semiconductor packaging projects.
- Collaborating with cross-functional design and process teams on package-level signal/power integrity considerations.

PROJECTS

Signal Integrity & Power Integrity Analysis using Ansys HFSS

- Performed SI/PI analysis of **PCIe Gen4** differential pairs, microstrip, stripline, and silicon interposer structures for HBM2E systems.
- Extracted RLC parameters of advanced electronic packages, including BGA and wire-bond structures.
- Analysed S-parameters, insertion loss, return loss, and crosstalk (NEXT/FEXT); designed a U-shaped microstrip to observe return path variations at high and low frequencies.

EM Simulation and VNA-Based Validation of Microstrip Transmission Lines (Ansys HFSS, VNA)

- Designed FR4 microstrip transmission lines with different bends and via dimensions
- Extracted S-parameters through simulation in Ansys HFSS
- Correlated simulated and measured VNA results to validate behaviour
- Identified practical design effects arising from layout variations

Temperature Monitoring System using Altium Designer

- Designed a DA14531 MCU-based circuit with a TMP102 I2C temperature sensor, including filtering, decoupling, and pull-up networks.
- Created schematic, PCB layout, and fabrication drafts in Altium Designer, ensuring proper grounding and signal integrity. Added an LED over-temperature indicator and verified the design with ERC/DRC checks for manufacturability.

RELEVANT COURSES

- **IC Packaging Domain:** Electronic Package Design, Electronic Packaging Materials, Heat Transfer in Electronic Packaging, Fatigue and Fracture in Electronic Packaging, IC Assembly, Packaging Design and Characterisation Lab
- **Industry-Centric Courses:** Fundamentals of **IC Packaging Assembly and Manufacturing (Micron)**. Gained knowledge in semiconductor packaging, testing, and reliability. Covered end-to-end processes, including die preparation, flip-chip, backend test flows, failure analysis, and quality systems like FMEA and SPC used in advanced semiconductor manufacturing. PCB design and layout for electronic systems (Renesas). Designed multi-layer PCBs using **Altium Designer** with a focus on schematic capture, routing, and DRC, with emphasis on signal integrity, manufacturability, and hardware reliability aligned with industry standards.
- **Analog Domain:** Analog IC Design, Analog IC Design Lab, RF IC Design.
- **Fabrication Domain:** VLSI Technology.

CONFERENCES AND WORKSHOPS

Indian Semiconductors & Packaging Ecosystem Conference 2025

- Presented a poster titled "Design and Demonstrate Low-Temperature Fine-Pitch (less than 2mm) Cu-Cu Bonding" under Prof. Shiv Govind Singh (IITH), Dr Naresh Emani (IITH), Dr Nilesh Badwe (IITK).
- Focused on developing and demonstrating advanced 2.5D/3D integration technologies.
- Key research: fabrication of fine-pitch, high-aspect-ratio through-silicon vias (TSVs) and investigation of low-temperature Cu-Cu hybrid bonding for high-density packaging.
- Expertise in Si-Si bonding, interface property analysis, and wafer-scale electrical performance evaluation.

ACHIEVEMENTS

- Presented a poster titled **"Design and Demonstrate Low-Temperature Fine-Pitch (less than 2mm) Cu-Cu Bonding"** at the ISPEC Conference, Gandhinagar, 2025.
- Achieved 88.78 percentile in GATE 2023 (EC), reflecting strong core fundamentals.

EXTRACURRICULAR & POSITION OF RESPONSIBILITIES

Cricket Team Player

- Represented the college team and finished as runner-up in the **Inter-IIT Cricket** tournament at IITH.
- Played a one-day match for the club under the **Hyderabad Cricket Association (HCA)**.
- Managed event photography and promotional footage for club activities, enhancing visibility across college .

TECHNICAL SKILLS

Signal & Power Integrity: PDN modelling (die-to-package-to-board), impedance-controlled routing, S-parameter characterisation, voltage-droop/ground-bounce analysis, eye diagram validation, HFSS-VNA/TDR correlation

Simulation & EDA Tools: Ansys HFSS, Ansys Q3D Extractor, Ansys Icepak, Cadence Virtuoso, Cadence Spectre, Cadence Allegro, LTSpice, Altium Designer, Xilinx Vivado. Languages & Scripting: Verilog, MATLAB, Python, C

Packaging & Reliability: 2.5D/3D interconnect (CoWoS, TSV, Cu-Cu hybrid bonding), thermo-mechanical reliability, CTE mismatch analysis, BGA/wire-bond parasitic extraction

Analog / Mixed-Signal IC Design: CMOS amplifiers, current mirrors, bandgap references, mixed-signal DAC design, Monte Carlo mismatch analysis in Cadence Virtuoso